Comprehensive Mathematics Reference

Volume 1: Basic Mathematics

From Absolute Beginner to Confident Practitioner

Structured for Learning, Teaching, and Retrieval-Augmented Generation (RAG) Systems

2026

About This Volume

This volume is the first in a planned multi-volume comprehensive mathematics reference, beginning with **Basic Mathematics** and covering: Numbers, Arithmetic, Fractions, Decimals, Percentages, Ratios, Proportions, and Simplification.

Each section follows a consistent, predictable structure so that content is easy to study sequentially *and* easy to retrieve accurately by automated systems (RAG pipelines, search indexes, tutoring bots):

- **Beginner Explanation** — plain-language introduction assuming no prior knowledge
- **Intermediate Explanation** — deeper theory and mechanics
- **Advanced Explanation** — rigor, edge cases, and connections to later topics
- **Key Definitions** — precise, quotable definitions
- **Formula Sheet** — all formulas for the section in one place
- **Worked Examples** — fully solved, step-by-step
- **Practice Questions** — with detailed solutions
- **Multiple-Choice Questions** — with answers and explanations
- **Real-World Applications**
- **Common Mistakes**
- **Interview-Style Conceptual Questions**
- **FAQs**

Terminology is kept consistent across the entire volume: once a term is defined, later sections reuse that exact definition rather than restating it differently, which reduces contradiction risk for retrieval systems. Every subsection has a descriptive, unique heading suitable for use as a retrieval chunk title.

Note on scope: this volume is intentionally complete and self-contained for Basic Mathematics rather than padded with filler. Subsequent volumes (Algebra, Geometry, Trigonometry, Calculus, Linear Algebra, Probability & Statistics, Discrete Mathematics, and advanced/applied topics through Machine Learning Mathematics) follow the same structural template and can be produced as separate volumes and later merged.

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Chapter 1

Numbers

1.1 Introduction to Numbers

1.1.1 Beginner Explanation

A **number** is a mathematical object used to count, measure, and label. Numbers are the foundation of all mathematics — every other topic in this book, from algebra to calculus, is built using numbers and the rules for combining them.

The most familiar numbers are the ones used for counting: $1, 2, 3, 4, 5, \ldots$. These are called **natural numbers**. As mathematics developed, more types of numbers were invented to solve problems that natural numbers alone could not handle — for example, representing "nothing" (zero), amounts owed (negative numbers), parts of a whole (fractions), and quantities that cannot be written as exact fractions (irrational numbers like π).

1.1.2 Intermediate Explanation

Numbers are organized into nested categories called **number systems**. Each system extends the previous one to solve equations or represent quantities that the smaller system cannot handle.

System	Symbol	Example Elements
Natural Numbers	$\mathbb{N}$	$1, 2, 3, 4, \ldots$
Whole Numbers	$\mathbb{W}$	$0, 1, 2, 3, \ldots$
Integers	$\mathbb{Z}$	$\ldots, -2, -1, 0, 1, 2, \ldots$
Rational Numbers	$\mathbb{Q}$	$\frac{1}{2}, -\frac{3}{4}, 5, 0.75$
Irrational Numbers	—	$\pi, \sqrt{2}, e$
Real Numbers	$\mathbb{R}$	All rational and irrational numbers
Complex Numbers	$\mathbb{C}$	$3 + 4i$

Each system is a strict superset of the one before it (except irrational numbers, which are disjoint from rational numbers, but together with them form the real numbers):

$$\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

1.1.3 Advanced Explanation

Formally, each number system can be constructed rigorously from a simpler one:

- $\mathbb{N}$ can be built from set theory using the **Peano axioms**, which define 0 (or 1) and a "successor" operation.
- $\mathbb{Z}$ is constructed from $\mathbb{N}$ as equivalence classes of ordered pairs (a, b) representing $a - b$, which allows subtraction to always be defined.

- $\mathbb{Q}$ is constructed from $\mathbb{Z}$ as equivalence classes of ordered pairs (p, q) with $q \neq 0$, representing $\frac{p}{q}$, which allows division (except by zero) to always be defined.
- $\mathbb{R}$ is constructed from $\mathbb{Q}$ via **Dedekind cuts** or **Cauchy sequences**, which "fill the gaps" left by irrational numbers so that every bounded set has a least upper bound (completeness property).
- $\mathbb{C}$ is constructed from $\mathbb{R}$ by adjoining $i = \sqrt{-1}$, giving *algebraic closure*: every polynomial equation with complex coefficients has a complex root (the Fundamental Theorem of Algebra).

This chain of constructions is a recurring pattern in mathematics: extend a system just enough to make a previously unsolvable operation solvable, while preserving all the rules that held before.

1.1.4 Key Definitions

Definition: Natural Number

A natural number is one of the counting numbers $1, 2, 3, 4, \ldots$ (some conventions include 0; this book uses $\mathbb{N} = \{1, 2, 3, \ldots\}$ and $\mathbb{W} = \{0, 1, 2, 3, \ldots\}$ for whole numbers).

Definition: Integer

An integer is any natural number, its negative, or zero: $\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$.

Definition: Rational Number

A rational number is any number that can be written as a fraction $\frac{p}{q}$, where p and q are integers and $q \neq 0$.

Definition: Irrational Number

An irrational number is a real number that *cannot* be written as a fraction of two integers. Its decimal expansion is infinite and never repeats.

Definition: Real Number

A real number is any number on the number line — the union of all rational and irrational numbers.

Definition: Prime Number

A prime number is a natural number greater than 1 whose only positive divisors are 1 and itself.

Definition: Composite Number

A composite number is a natural number greater than 1 that is not prime (i.e., it has at least one divisor other than 1 and itself).

Definition: Even and Odd Numbers

An integer n is *even* if $n = 2k$ for some integer k, and *odd* if $n = 2k + 1$ for some integer k.

1.1.5 Formula Sheet

Formula: Numbers

- Absolute value: $|x| = x$ if $x \geq 0$; $|x| = -x$ if $x < 0$.
- Number of integers between a and b inclusive: $b - a + 1$.
- Sum of first n natural numbers: $1 + 2 + \cdots + n = \dfrac{n(n+1)}{2}$.
- A number n is even $\iff n \bmod 2 = 0$; odd $\iff n \bmod 2 = 1$.

1.1.6 Worked Examples

Worked Example: Classifying a Number

Problem: Classify -7 according to the number systems it belongs to.
Solution:

1. Is it a natural number? No — natural numbers are positive $(1, 2, 3, \dots)$, and -7 is negative.
2. Is it a whole number? No — whole numbers are $0, 1, 2, \dots$, all non-negative.
3. Is it an integer? Yes — it is a whole negative number.
4. Is it rational? Yes — every integer n can be written as $\frac{n}{1}$, so $-7 = \frac{-7}{1}$.
5. Is it real? Yes — all rational numbers are real.

Answer: -7 is an integer, a rational number, and a real number, but not a natural or whole number.

Worked Example: Identifying Primes

Problem: List all prime numbers between 1 and 30.
Solution: Use the Sieve of Eratosthenes: list numbers 2 to 30, and cross out multiples of each prime starting from 2.

- Cross out multiples of 2 (except 2): $4, 6, 8, \dots, 30$.
- Cross out multiples of 3 (except 3): $9, 15, 21, 27$ remain to remove.
- Cross out multiples of 5 (except 5): 25 remains to remove.
- $\sqrt{30} \approx 5.48$, so we only need to check divisibility up to 5.

Answer: $2, 3, 5, 7, 11, 13, 17, 19, 23, 29$.

Worked Example: Rational or Irrational?

Problem: Determine whether $\sqrt{16}$ and $\sqrt{17}$ are rational or irrational.
Solution:

- $\sqrt{16} = 4$, since $4 \times 4 = 16$. Since $4 = \frac{4}{1}$, it is rational.
- 17 is not a perfect square (no integer squared equals 17). The square root of any

non-perfect-square natural number is irrational. So $\sqrt{17}$ is irrational.

Answer: $\sqrt{16} = 4$ is rational; $\sqrt{17}$ is irrational.

Worked Example: Absolute Value

Problem: Evaluate $|-9| + |5| - |-2|$.
Solution:

$$|-9| = 9, \quad |5| = 5, \quad |-2| = 2$$

$$9 + 5 - 2 = 12$$

Answer: 12.

1.1.7 Practice Questions

Q1. Is 0 a natural number, whole number, integer, rational number, and/or real number? (Use the convention $\mathbb{N} = \{1, 2, 3, \dots\}$.)

Q2. Classify $\frac{5}{2}$ into all number systems it belongs to.

Q3. List all composite numbers between 20 and 40.

Q4. Is π rational or irrational? Explain why.

Q5. Evaluate $|-12| - |7| + |-3|$.

Q6. True or False: every integer is a rational number.

Q7. True or False: every rational number is an integer.

Q8. Is $0.333\dots$ (repeating) rational or irrational?

Q9. Find the sum of the first 10 natural numbers using the formula.

Q10. Is -4 even or odd? Justify using the definition.

Detailed Solutions

A1. 0 is not a natural number under this convention, but it *is* a whole number, an integer, a rational number ($\frac{0}{1}$), and a real number.

A2. $\frac{5}{2} = 2.5$ is not a natural number, whole number, or integer (it is not a whole quantity). It *is* rational (it is literally written as a ratio of integers) and real.

A3. Composite numbers between 20 and 40: $21, 22, 24, 25, 26, 27, 28, 30, 32, 33, 34, 35, 36, 38, 39$ (all numbers in range except the primes $23, 29, 31, 37$).

A4. π is irrational. It cannot be expressed as a ratio of two integers, and its decimal expansion is infinite and non-repeating. (This is a proven mathematical fact, first shown by Johann Lambert in 1761.)

A5. $|-12| - |7| + |-3| = 12 - 7 + 3 = 8$.

A6. **True.** Any integer n can be written as $\frac{n}{1}$, satisfying the definition of a rational number.

A7. **False.** Counterexample: $\frac{1}{2}$ is rational but not an integer.

A8. **Rational.** $0.\overline{3} = \frac{1}{3}$, a ratio of two integers. Any repeating decimal is rational.

A9. $\dfrac{n(n+1)}{2} = \dfrac{10 \times 11}{2} = \dfrac{110}{2} = 55.$

A10. **Even.** By definition, n is even if $n = 2k$ for some integer k. Here $-4 = 2 \times (-2)$, and -2 is an integer, so -4 is even.

1.1.8 Multiple-Choice Questions

MCQ1. Which of the following is *not* a rational number?

(a) $\frac{3}{7}$
(b) -5
(c) $\sqrt{2}$
(d) 0.25

Answer: (c) $\sqrt{2}$ is irrational; the others can each be written as a ratio of integers.

MCQ2. Which set contains -3?

(a) Natural numbers only
(b) Whole numbers only
(c) Integers, rational numbers, and real numbers
(d) None of these

Answer: (c).

MCQ3. How many prime numbers are there between 1 and 10?

(a) 2
(b) 3
(c) 4
(d) 5

Answer: (c) — $2, 3, 5, 7$.

MCQ4. Which of the following is always true?

(a) Every whole number is a natural number.
(b) Every natural number is a whole number.
(c) Every rational number is an integer.
(d) Every real number is rational.

Answer: (b).

1.1.9 Real-World Applications

Real-World Application: Number Systems in Computing

Computers represent all data using binary (base-2) integers at the hardware level. Understanding integers and their properties (even/odd, sign) is essential for tasks like memory addressing, error-checking (parity bits), and cryptography, where prime numbers are the foundation of encryption algorithms such as RSA.

Real-World Application: Irrational Numbers in Engineering

The number π (irrational) is essential in any calculation involving circles or rotation — from designing gears and wheels to calculating orbital paths of satellites. Engineers use rational approximations (like 3.14159) since irrational numbers cannot be written exactly in decimal form, understanding that this introduces a small, quantifiable rounding error.

1.1.10 Common Mistakes

Common Mistake: Confusing Whole Numbers and Natural Numbers

Students often use "whole number" and "natural number" interchangeably. In most conventions, whole numbers include 0 while natural numbers (in this text) start at 1. Always check which convention a textbook or exam is using.

Common Mistake: Assuming All Square Roots Are Irrational

A common error is assuming $\sqrt{n}$ is always irrational. This is false when n is a perfect square: $\sqrt{25} = 5$ is rational (in fact, an integer). Only non-perfect-square natural numbers have irrational square roots.

Common Mistake: Forgetting the Sign in Absolute Value

$|-x|$ is not $-x$ in general — it is the non-negative value. A frequent mistake is writing $|-5| = -5$ instead of the correct $|-5| = 5$.

1.1.11 Interview-Style Conceptual Questions

Conceptual / Interview Question: Why do we need irrational numbers at all?

Answer: Because the rational numbers, despite being infinite and dense (between any two rationals there's another rational), still leave "gaps" on the number line. For example, no rational number squared equals exactly 2. Without irrational numbers, basic geometric quantities like the diagonal of a unit square ($\sqrt{2}$) could not be represented as a number at all. Irrational numbers fill these gaps, making the real number line "complete" — every point on the line corresponds to a real number.

Conceptual / Interview Question: Is zero even or odd?

Answer: Zero is even. By definition, an integer n is even if it can be written as $n = 2k$ for some integer k. Since $0 = 2 \times 0$ and 0 is an integer, zero satisfies the definition of even.

1.1.12 FAQs

FAQ: Is 0 a positive or negative number?

Neither. Zero is neither positive nor negative; it is the boundary between them on the number line.

FAQ: Are all fractions rational numbers?

Yes, provided the numerator and denominator are integers and the denominator is nonzero. A "fraction" like $\frac{\sqrt{2}}{3}$ is not rational, since $\sqrt{2}$ is not an integer.

FAQ: Why is division by zero undefined?

Division $\frac{a}{b}$ is defined as the number x such that $b \times x = a$. If $b = 0$, then $0 \times x = 0$ for every x, so there is no unique answer (or, if $a \neq 0$, no answer exists at all). This is why $\frac{a}{0}$ is undefined for all a.

1.2 Place Value and the Decimal System

1.2.1 Beginner Explanation

The decimal (base-10) number system represents numbers using ten digits, 0 through 9, where the position of each digit determines its value. This is called **place value**.

For example, in the number *presentlove* : 3,482:

- 3 is in the thousands place, worth 3000
- 4 is in the hundreds place, worth 400
- 8 is in the tens place, worth 80
- 2 is in the ones (units) place, worth 2

So $3482 = 3000 + 400 + 80 + 2$.

1.2.2 Intermediate Explanation

Formally, any whole number N in base 10 can be written in **expanded form**:

$$N = d_n \times 10^n + d_{n-1} \times 10^{n-1} + \cdots + d_1 \times 10^1 + d_0 \times 10^0$$

where each digit $d_i \in \{0, 1, \ldots, 9\}$. Moving one place to the left multiplies the place value by 10; moving one place to the right divides it by 10. This is why multiplying by 10 simply appends a zero, and dividing by 10 shifts the decimal point one place left.

1.2.3 Advanced Explanation

Decimal is one specific case of a general **positional numeral system** with base (radix) b. In base b, digits range from 0 to $b-1$, and place values are powers of b. Other common bases used in computing include binary ($b = 2$, digits $0, 1$), octal ($b = 8$), and hexadecimal ($b = 16$, digits 0–9 and A–F). Converting between bases relies on the same expanded-form idea, just with a different base for the powers.

1.2.4 Key Definitions

Definition: Place Value

The value a digit represents based on its position within a number.

Definition: Face Value

The value of a digit itself, regardless of its position (e.g., in 482, the face value of 8 is simply 8).

Definition: Expanded Form

Writing a number as the sum of the place values of each digit.

1.2.5 Formula Sheet

Formula: Place Value

$$N = \sum_{i=0}^{n} d_i \cdot 10^i, \qquad d_i \in \{0, \ldots, 9\}$$

Multiplying by 10^k: shift digits k places left (append k zeros for integers).
Dividing by 10^k: shift digits k places right (move decimal point k places left).

1.2.6 Worked Examples

Worked Example: Expanded Form

Problem: Write 50,726 in expanded form.
Solution:

$$\begin{aligned} 50726 &= 5 \times 10^4 + 0 \times 10^3 + 7 \times 10^2 + 2 \times 10^1 + 6 \times 10^0 \\ &= 50000 + 0 + 700 + 20 + 6 \end{aligned}$$

Answer: $50000 + 700 + 20 + 6$.

Worked Example: Place Value vs Face Value

Problem: In the number 6,349, find the place value and face value of the digit 3.
Solution: The digit 3 is in the hundreds position.

- Face value $= 3$
- Place value $= 3 \times 100 = 300$

Answer: Face value 3; place value 300.

1.2.7 Practice Questions

Q1. Write 709,214 in expanded form.

Q2. Find the place value of 7 in 27,841.

Q3. What is $4{,}500 \times 100$?

Q4. What is $8{,}200 \div 100$?

Q5. Write the number represented by $7 \times 10^3 + 2 \times 10^2 + 0 \times 10^1 + 9 \times 10^0$.

Detailed Solutions

A1. $709214 = 7\times10^5+0\times10^4+9\times10^3+2\times10^2+1\times10^1+4\times10^0 = 700000+9000+200+10+4$.

A2. The 7 is in the thousands place: place value $= 7000$.

A3. Multiplying by 100 shifts two places left: $4500 \times 100 = 450000$.

A4. Dividing by 100 shifts two places right: $8200 \div 100 = 82$.

A5. $7000 + 200 + 0 + 9 = 7209$.

1.2.8 Real-World Applications

Real-World Application: Currency and Accounting

Place value underlies all financial record-keeping — the difference between \$1,000 and \$10,000 is a single misplaced digit, which is why accounting systems use strict place-value formatting and digit grouping (commas) to prevent costly errors.

1.2.9 Common Mistakes

Common Mistake: Miscounting Zeros When Multiplying by Powers of 10

A frequent error is adding the wrong number of zeros, e.g., computing 340×1000 as 34000 instead of 340000. Always count the zeros in the power of ten carefully, or use exponent form: 340×10^3.

1.2.10 Interview-Style Conceptual Questions

Conceptual / Interview Question: Why does the position of a digit matter more than the digit itself?

Answer: Because the decimal system is positional — the same digit represents different quantities depending on where it sits. This is what allows just 10 symbols (0–9) to represent infinitely many numbers efficiently, unlike systems (e.g., Roman numerals) where new symbols are needed for larger values.

1.2.11 FAQs

FAQ: Why is it called "base 10"?

Because there are 10 digits (0–9), and each place value is a power of 10. "Base" refers to the number of unique digits used before the system "carries over" to a new place.

1.3 Divisibility, Factors, and Multiples

1.3.1 Beginner Explanation

A number a is **divisible** by a number b if dividing a by b leaves no remainder. For example, 12 is divisible by 3 because $12 \div 3 = 4$ exactly. A **factor** (or divisor) of a number is any number that divides it evenly. A **multiple** of a number is what you get by multiplying it by an integer.

1.3.2 Intermediate Explanation

Every natural number greater than 1 can be broken down into prime factors — this is called **prime factorization**. Because of the **Fundamental Theorem of Arithmetic**, this factorization is unique (up to the order of factors). Prime factorization is the basis for finding the **Greatest Common Divisor (GCD)** and **Least Common Multiple (LCM)** of two or more numbers.

1.3.3 Advanced Explanation

The Fundamental Theorem of Arithmetic states that every integer $n > 1$ can be written uniquely as

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

where $p_1 < p_2 < \cdots < p_k$ are primes and $a_i \geq 1$. This underlies the Euclidean Algorithm for computing GCDs efficiently (without full factorization), which runs in $O(\log(\min(a, b)))$ steps and is foundational to modern cryptographic systems such as RSA.

1.3.4 Key Definitions

Definition: Divisibility

An integer a is divisible by a nonzero integer b if there exists an integer k such that $a = bk$.

Definition: Factor (Divisor)

A factor of n is a positive integer that divides n with no remainder.

Definition: Multiple

A multiple of n is any number of the form $n \times k$ for an integer k.

Definition: Greatest Common Divisor (GCD)

The largest positive integer that divides two or more given numbers without remainder.

Definition: Least Common Multiple (LCM)

The smallest positive integer that is a multiple of two or more given numbers.

1.3.5 Formula Sheet

Formula: Divisibility Rules

- Divisible by 2: last digit is even $(0, 2, 4, 6, 8)$.
- Divisible by 3: sum of digits divisible by 3.
- Divisible by 4: last two digits form a number divisible by 4.
- Divisible by 5: last digit is 0 or 5.
- Divisible by 6: divisible by both 2 and 3.
- Divisible by 9: sum of digits divisible by 9.
- Divisible by 10: last digit is 0.

- $\text{GCD}(a, b) \times \text{LCM}(a, b) = a \times b$.
- Euclidean Algorithm: $\text{GCD}(a, b) = \text{GCD}(b, a \bmod b)$, until remainder is 0.

1.3.6 Worked Examples

Worked Example: Prime Factorization

Problem: Find the prime factorization of 360.
Solution:

$$360 = 2 \times 180 = 2 \times 2 \times 90 = 2 \times 2 \times 2 \times 45 = 2^3 \times 45 = 2^3 \times 3^2 \times 5$$

Answer: $360 = 2^3 \times 3^2 \times 5$.

Worked Example: GCD Using Euclidean Algorithm

Problem: Find $\text{GCD}(252, 105)$.
Solution:

$$252 = 2 \times 105 + 42$$

$$105 = 2 \times 42 + 21$$

$$42 = 2 \times 21 + 0$$

The last nonzero remainder is 21.
Answer: $\text{GCD}(252, 105) = 21$.

Worked Example: LCM Using GCD Formula

Problem: Find $\text{LCM}(252, 105)$.
Solution: Using $\text{GCD}(a, b) \times \text{LCM}(a, b) = a \times b$ and GCD $= 21$ from the previous example:

$$\text{LCM}(252, 105) = \frac{252 \times 105}{21} = \frac{26460}{21} = 1260$$

Answer: 1260.

1.3.7 Practice Questions

Q1. Find the prime factorization of 180.

Q2. Is 273 divisible by 3? Use the divisibility rule to check.

Q3. Find $\text{GCD}(48, 18)$ using the Euclidean Algorithm.

Q4. Find $\text{LCM}(12, 18)$.

Q5. List all factors of 36.

Q6. Two bells ring every 18 and 24 minutes respectively, starting together. After how many minutes will they ring together again?

Detailed Solutions

A1. $180 = 2 \times 90 = 2 \times 2 \times 45 = 2^2 \times 45 = 2^2 \times 3^2 \times 5$.

A2. Sum of digits: $2 + 7 + 3 = 12$, and 12 is divisible by 3. So yes, 273 is divisible by 3 ($273 = 3 \times 91$).

A3. $48 = 2 \times 18 + 12$; $18 = 1 \times 12 + 6$; $12 = 2 \times 6 + 0$. GCD $= 6$.

A4. Using GCD$(12, 18)$: $18 = 1 \times 12 + 6$; $12 = 2 \times 6 + 0$, so GCD$= 6$. Then LCM $= \frac{12 \times 18}{6} = \frac{216}{6} = 36$.

A5. Factors of 36: $1, 2, 3, 4, 6, 9, 12, 18, 36$.

A6. This is an LCM problem: LCM$(18, 24)$. $18 = 2 \times 3^2$, $24 = 2^3 \times 3$. LCM takes the highest power of each prime: $2^3 \times 3^2 = 8 \times 9 = 72$. **They ring together again after 72 minutes.**

1.3.8 Multiple-Choice Questions

MCQ1. Which of the following is a factor of 84?

(a) 9

(b) 14

(c) 25

(d) 32

Answer: (b) — $84 = 14 \times 6$.

MCQ2. What is GCD$(0, 15)$?

(a) 0

(b) 1

(c) 15

(d) Undefined

Answer: (c) — by convention, GCD$(0, n) = n$ since every number divides 0.

1.3.9 Real-World Applications

Real-World Application: Scheduling Problems

LCM is used to solve recurring-event scheduling problems, such as determining when two people's work shifts will next align, or when repeating traffic light cycles will next synchronize.

Real-World Application: Cryptography

The GCD (via the Euclidean Algorithm) and prime factorization are central to RSA encryption, which relies on the practical difficulty of factoring the product of two very large primes.

1.3.10 Common Mistakes

Common Mistake: Confusing Factors and Multiples

Students often mix up "factors of n" (numbers that divide n, always $\leq n$) with "multiples of n" (numbers n divides into, always $\geq n$). Remember: factors are few and $\leq n$; multiples are infinite and $\geq n$.

Common Mistake: Using GCD Formula Incorrectly for Three Numbers

$\text{GCD}(a, b) \times \text{LCM}(a, b) = a \times b$ only works for *two* numbers. For three or more numbers, GCD and LCM must be computed pairwise/iteratively, not with this direct formula.

1.3.11 Interview-Style Conceptual Questions

Conceptual / Interview Question: Why does the Euclidean Algorithm work?

Answer: It relies on the fact that $\text{GCD}(a, b) = \text{GCD}(b, a \bmod b)$. This is true because any common divisor of a and b must also divide $a - qb$ for any integer q (including the remainder when a is divided by b), and vice versa. By repeatedly replacing the larger number with the remainder, the pair shrinks until the remainder is 0, at which point the other number is the GCD.

1.3.12 FAQs

FAQ: Is 1 a prime number?

No. By definition, a prime number has exactly two distinct positive divisors: 1 and itself. The number 1 has only one positive divisor (itself), so it is neither prime nor composite.

FAQ: What is the GCD of two prime numbers?

If the two primes are different, their GCD is 1 (they share no common factors other than 1). If they are the same prime, the GCD is that prime itself.

Chapter 2

Arithmetic

2.1 The Four Basic Operations

2.1.1 Beginner Explanation

Arithmetic is built on four fundamental operations: **addition**, **subtraction**, **multiplication**, and **division**. These operations let us combine numbers to find totals, differences, products, and shares.

- **Addition** ($+$): combining two quantities into a total. $3 + 4 = 7$.
- **Subtraction** ($-$): finding the difference between quantities. $7 - 3 = 4$.
- **Multiplication** ($\times$): repeated addition. $4 \times 3 = 4 + 4 + 4 = 12$.
- **Division** ($\div$): splitting into equal parts. $12 \div 4 = 3$.

2.1.2 Intermediate Explanation

Each operation has an **inverse**: subtraction undoes addition, and division undoes multiplication. Operations also have important structural properties:

- **Commutative Property**: order doesn't matter. $a + b = b + a$ and $a \times b = b \times a$. (Subtraction and division are *not* commutative: $5 - 3 \neq 3 - 5$.)
- **Associative Property**: grouping doesn't matter. $(a+b)+c = a+(b+c)$ and $(a \times b) \times c = a \times (b \times c)$.
- **Distributive Property**: multiplication distributes over addition. $a \times (b+c) = a \times b + a \times c$.
- **Identity Elements**: 0 is the additive identity ($a + 0 = a$); 1 is the multiplicative identity ($a \times 1 = a$).
- **Inverse Elements**: $-a$ is the additive inverse of a ($a + (-a) = 0$); $\frac{1}{a}$ is the multiplicative inverse of a (for $a \neq 0$), since $a \times \frac{1}{a} = 1$.

2.1.3 Advanced Explanation

These properties define $(\mathbb{R}, +, \times)$ as a mathematical structure called a **field**: a set with two operations that satisfy commutativity, associativity, distributivity, and have identity and inverse elements (except 0 has no multiplicative inverse). Recognizing this structure is what allows algebra to generalize arithmetic — the same rules that govern numbers govern variables, matrices (under certain conditions), polynomials, and more, wherever a field or ring structure applies.

2.1.4 Key Definitions

Definition: Sum, Difference, Product, Quotient

The *sum* is the result of addition; the *difference* is the result of subtraction; the *product* is the result of multiplication; the *quotient* is the result of division.

Definition: Order of Operations (PEMDAS/BODMAS)

The convention that determines the sequence in which operations in an expression are evaluated: Parentheses/Brackets, Exponents/Orders, Multiplication and Division (left to right), Addition and Subtraction (left to right).

2.1.5 Formula Sheet

Formula: Arithmetic Properties

- Commutative: $a + b = b + a; \quad ab = ba$
- Associative: $(a + b) + c = a + (b + c); \quad (ab)c = a(bc)$
- Distributive: $a(b + c) = ab + ac$
- Identity: $a + 0 = a; \quad a \times 1 = a$
- Inverse: $a + (-a) = 0; \quad a \times \frac{1}{a} = 1 \ (a \neq 0)$
- Order of Operations: Parentheses $\rightarrow$ Exponents $\rightarrow$ Multiplication/Division (L to R) $\rightarrow$ Addition/Subtraction (L to R)

2.1.6 Worked Examples

Worked Example: Order of Operations

Problem: Evaluate $6 + 4 \times (5 - 2)^2 \div 3$.
Solution:

$$\begin{aligned} 6 + 4 \times (5 - 2)^2 \div 3 &= 6 + 4 \times 3^2 \div 3 \quad \text{(parentheses)} \\ &= 6 + 4 \times 9 \div 3 \quad \text{(exponent)} \\ &= 6 + 36 \div 3 \quad \text{(multiplication)} \\ &= 6 + 12 \quad \text{(division)} \\ &= 18 \quad \text{(addition)} \end{aligned}$$

Answer: 18.

Worked Example: Using the Distributive Property

Problem: Compute 23×99 using the distributive property (mental math shortcut).
Solution: Rewrite $99 = 100 - 1$:

$$23 \times 99 = 23 \times (100 - 1) = 23 \times 100 - 23 \times 1 = 2300 - 23 = 2277$$

Answer: 2277.

Worked Example: Long Division

Problem: Divide $6825 \div 25$.
Solution:

- $68 \div 25 = 2$ remainder 18 (since $25 \times 2 = 50$, $68 - 50 = 18$)
- Bring down 2: $182 \div 25 = 7$ remainder 7 (since $25 \times 7 = 175$, $182 - 175 = 7$)
- Bring down 5: $75 \div 25 = 3$ remainder 0

Answer: $6825 \div 25 = 273$.

2.1.7 Practice Questions

Q1. Evaluate $12 - 3 \times (2 + 1)^2 \div 9$.

Q2. Compute 47×102 using the distributive property.

Q3. Divide $4536 \div 18$.

Q4. Which property justifies $5 \times (2 + 7) = 5 \times 2 + 5 \times 7$?

Q5. Evaluate $(8 - 3) \times (6 \div 2) + 4$.

Detailed Solutions

A1. $12 - 3 \times (3)^2 \div 9 = 12 - 3 \times 9 \div 9 = 12 - 27 \div 9 = 12 - 3 = 9$.

A2. $47 \times 102 = 47 \times (100 + 2) = 4700 + 94 = 4794$.

A3. $4536 \div 18$: $45 \div 18 = 2$ r9; bring down 3: $93 \div 18 = 5$ r3; bring down 6: $36 \div 18 = 2$ r0. Answer: 252.

A4. The Distributive Property.

A5. $(5) \times (3) + 4 = 15 + 4 = 19$.

2.1.8 Real-World Applications

Real-World Application: Budgeting and Everyday Finance

The four basic operations underlie all budgeting: addition for totaling expenses, subtraction for tracking remaining balance, multiplication for scaling costs (e.g., price per item $\times$ quantity), and division for splitting shared bills.

2.1.9 Common Mistakes

Common Mistake: Ignoring Order of Operations

A very common error is evaluating left-to-right without respecting precedence, e.g., computing $2 + 3 \times 4$ as $(2 + 3) \times 4 = 20$ instead of the correct $2 + (3 \times 4) = 14$.

Common Mistake: Assuming Subtraction and Division Are Commutative

Unlike addition and multiplication, $a - b \neq b - a$ and $a \div b \neq b \div a$ in general. Always preserve the order given in the problem.

2.1.10 Interview-Style Conceptual Questions

Conceptual / Interview Question: Why is order of operations necessary at all?

Answer: Without an agreed convention, the same expression could be evaluated multiple different ways, giving different answers depending on who reads it. Order of operations is a universal grammar for mathematical expressions, ensuring $2 + 3 \times 4$ means the same thing to every mathematician, engineer, and computer program worldwide.

2.1.11 FAQs

FAQ: Does multiplication always come before division?

No. Multiplication and division have equal precedence and are evaluated left to right, in the order they appear — not "M before D."

2.2 Integers: Operations with Positive and Negative Numbers

2.2.1 Beginner Explanation

Integers include positive numbers, negative numbers, and zero. Adding and subtracting negative numbers can feel confusing at first, but a simple mental model helps: think of a number line, where positive numbers move right and negative numbers move left.

2.2.2 Intermediate Explanation

Key rules for signed number operations:

- Adding a negative is the same as subtracting: $a + (-b) = a - b$.
- Subtracting a negative is the same as adding: $a - (-b) = a + b$.
- The product (or quotient) of two numbers with the *same* sign is positive.
- The product (or quotient) of two numbers with *different* signs is negative.

2.2.3 Advanced Explanation

These rules follow directly from field axioms. For instance, $(-1) \times (-1) = 1$ can be proven: starting from $(-1) \times (1 + (-1)) = (-1) \times 0 = 0$, and distributing gives $(-1) \times 1 + (-1) \times (-1) = 0$, i.e., $-1 + (-1) \times (-1) = 0$, so $(-1) \times (-1)$ must equal 1 to satisfy the equation. This shows the "negative times negative is positive" rule is not arbitrary — it is a logical consequence of the distributive property and additive inverses.

2.2.4 Key Definitions

Definition: Signed Number

A number classified as positive, negative, or zero.

Definition: Additive Inverse (Opposite)

The number that, when added to a, yields 0; denoted $-a$.

2.2.5 Formula Sheet

Formula: Sign Rules

- $(+) + (+) = (+)$; $(-) + (-) = (-)$
- $a - (-b) = a + b$; $a + (-b) = a - b$
- $(+) \times (+) = (+)$; $(-) \times (-) = (+)$; $(+) \times (-) = (-)$
- $(+) \div (+) = (+)$; $(-) \div (-) = (+)$; $(+) \div (-) = (-)$

2.2.6 Worked Examples

Worked Example: Adding and Subtracting Negatives

Problem: Evaluate $-8 + 3 - (-5)$.
Solution:

$$-8 + 3 - (-5) = -8 + 3 + 5 = -5 + 5 = 0$$

Answer: 0.

Worked Example: Multiplying Signed Numbers

Problem: Evaluate $(-4) \times (-3) \times (-2)$.
Solution: Multiply pairwise, tracking sign:

$$(-4) \times (-3) = 12 \quad \text{(same signs} \rightarrow \text{positive)}$$

$$12 \times (-2) = -24 \quad \text{(different signs} \rightarrow \text{negative)}$$

Answer: -24. (Note: an odd number of negative factors gives a negative result.)

2.2.7 Practice Questions

Q1. Evaluate $-15 - (-9) + 4$.

Q2. Evaluate $(-6) \times 5 \times (-2)$.

Q3. Evaluate $(-36) \div (-4)$.

Q4. Evaluate $-7 \times (-1) \times (-1) \times (-1)$.

Q5. A submarine is at -120 meters (below sea level). It rises 45 meters, then descends 30 meters. What is its final depth?

Detailed Solutions

A1. $-15 + 9 + 4 = -6 + 4 = -2$.

A2. $(-6) \times 5 = -30$; $-30 \times (-2) = 60$.

A3. $(-36) \div (-4) = 9$ (same signs $\rightarrow$ positive).

A4. Four negative factors $\rightarrow$ positive: $-7 \times (-1) = 7$; $7 \times (-1) = -7$; $-7 \times (-1) = 7$. Answer: 7.

A5. $-120 + 45 = -75$; $-75 - 30 = -105$. **Final depth: 105 meters below sea level.**

2.2.8 Real-World Applications

Real-World Application: Temperature and Elevation

Negative numbers naturally represent temperatures below zero, elevations below sea level, and financial debt. Arithmetic with signed numbers lets us compute changes in these quantities directly.

2.2.9 Common Mistakes

Common Mistake: Sign Errors When Removing Parentheses

A frequent mistake is writing $5 - (-3) = 5 - 3 = 2$ instead of the correct $5 - (-3) = 5 + 3 = 8$. Always convert "minus a negative" to "plus" before simplifying.

2.2.10 Interview-Style Conceptual Questions

Conceptual / Interview Question: Why is a negative times a negative positive?

Answer: It preserves the distributive property. If $(-1) \times (-1)$ were -1 instead of 1, then $(-1) \times (1 + (-1))$ would compute to two different values depending on whether you distribute first or add first — breaking consistency. The rule exists to keep arithmetic internally consistent, not as an arbitrary convention.

2.2.11 FAQs

FAQ: Is zero positive or negative?

Zero is neither positive nor negative — it is neutral, separating the two.

Chapter 3

Fractions

3.1 Understanding Fractions

3.1.1 Beginner Explanation

A **fraction** represents a part of a whole. It is written as $\frac{a}{b}$, where a (the **numerator**) tells you how many parts you have, and b (the **denominator**) tells you how many equal parts the whole is divided into. For example, $\frac{3}{4}$ means "3 out of 4 equal parts."

3.1.2 Intermediate Explanation

Fractions can be classified as:

- **Proper fraction**: numerator $<$ denominator (e.g., $\frac{2}{5}$), representing a value less than 1.
- **Improper fraction**: numerator $\geq$ denominator (e.g., $\frac{7}{4}$), representing a value ≥ 1.
- **Mixed number**: a whole number plus a proper fraction (e.g., $1\frac{3}{4}$), equivalent to an improper fraction.
- **Equivalent fractions**: different fractions representing the same value, e.g., $\frac{1}{2} = \frac{2}{4} = \frac{3}{6}$.

A fraction is in **lowest terms** (or **simplest form**) when the GCD of numerator and denominator is 1.

3.1.3 Advanced Explanation

Formally, a fraction $\frac{a}{b}$ represents the equivalence class of all pairs (a, b) related by $(a, b) \sim (c, d) \iff ad = bc$. This is precisely how the rational numbers $\mathbb{Q}$ are constructed from the integers $\mathbb{Z}$ (see Chapter 1). Every rational number has a unique representation in lowest terms with a positive denominator.

3.1.4 Key Definitions

Definition: Numerator and Denominator

In $\frac{a}{b}$, a is the numerator (parts counted) and b is the denominator (total equal parts; $b \neq 0$).

Definition: Lowest Terms

A fraction $\frac{a}{b}$ is in lowest terms if $\text{GCD}(a, b) = 1$.

Definition: Reciprocal

The reciprocal of $\frac{a}{b}$ (with $a \neq 0$) is $\frac{b}{a}$; the product of a fraction and its reciprocal is 1.

3.1.5 Formula Sheet

Formula: Fraction Conversions

- Mixed to improper: $n\frac{p}{q} = \dfrac{nq+p}{q}$
- Simplify: divide numerator and denominator by $\text{GCD}(a, b)$
- Reciprocal of $\frac{a}{b}$ is $\frac{b}{a}$

3.1.6 Worked Examples

Worked Example: Simplifying a Fraction

Problem: Simplify $\frac{48}{60}$.
Solution: $\text{GCD}(48, 60) = 12$.

$$\frac{48 \div 12}{60 \div 12} = \frac{4}{5}$$

Answer: $\frac{4}{5}$.

Worked Example: Mixed to Improper

Problem: Convert $3\frac{2}{7}$ to an improper fraction.
Solution:

$$3\frac{2}{7} = \frac{3 \times 7 + 2}{7} = \frac{23}{7}$$

Answer: $\frac{23}{7}$.

3.1.7 Practice Questions

Q1. Simplify $\frac{84}{126}$.

Q2. Convert $5\frac{3}{8}$ to an improper fraction.

Q3. Convert $\frac{29}{6}$ to a mixed number.

Q4. Are $\frac{6}{8}$ and $\frac{9}{12}$ equivalent? Show your work.

Detailed Solutions

A1. $\text{GCD}(84, 126) = 42$. $\frac{84 \div 42}{126 \div 42} = \frac{2}{3}$.

A2. $5\frac{3}{8} = \frac{5 \times 8 + 3}{8} = \frac{43}{8}$.

A3. $29 \div 6 = 4$ remainder 5, so $\frac{29}{6} = 4\frac{5}{6}$.

A4. Cross-multiply: $6 \times 12 = 72$ and $8 \times 9 = 72$. Equal, so **yes, they are equivalent**.

3.1.8 Common Mistakes

Common Mistake: Simplifying Only the Numerator or Denominator

A fraction must be divided by the *same* number on top and bottom to stay equivalent. Dividing only the numerator (or only the denominator) changes the fraction's value.

3.1.9 FAQs

FAQ: Can the denominator be zero?

No. Division by zero is undefined, so a fraction's denominator can never be 0.

3.2 Operations with Fractions

3.2.1 Beginner Explanation

To add or subtract fractions, they must have the **same denominator** (a "common denominator"). To multiply fractions, multiply numerators together and denominators together. To divide fractions, multiply by the reciprocal of the divisor.

3.2.2 Intermediate Explanation

When denominators differ, find the **Least Common Denominator (LCD)** — the LCM of the denominators — convert each fraction to an equivalent fraction with that denominator, then add or subtract numerators.

3.2.3 Advanced Explanation

Fraction arithmetic is a direct application of field operations on $\mathbb{Q}$:

$$\frac{a}{b}+\frac{c}{d}=\frac{ad+bc}{bd}, \qquad \frac{a}{b}\times\frac{c}{d}=\frac{ac}{bd}, \qquad \frac{a}{b}\div\frac{c}{d}=\frac{a}{b}\times\frac{d}{c}=\frac{ad}{bc}$$

Using the LCD instead of the raw product bd for addition/subtraction keeps numbers smaller but yields the same final simplified value.

3.2.4 Key Definitions

Definition: Least Common Denominator (LCD)

The LCM of the denominators of two or more fractions; the smallest common denominator they can share.

3.2.5 Formula Sheet

Formula: Fraction Operations

- Addition: $\frac{a}{b}+\frac{c}{d}=\frac{ad+bc}{bd}$
- Subtraction: $\frac{a}{b}-\frac{c}{d}=\frac{ad-bc}{bd}$
- Multiplication: $\frac{a}{b}\times\frac{c}{d}=\frac{ac}{bd}$

- Division: $\dfrac{a}{b} \div \dfrac{c}{d} = \dfrac{a}{b} \times \dfrac{d}{c} = \dfrac{ad}{bc}$

3.2.6 Worked Examples

Worked Example: Adding Fractions with Different Denominators

Problem: Evaluate $\frac{2}{3} + \frac{5}{6}$.
Solution: LCD of 3 and 6 is 6.

$$\frac{2}{3} = \frac{4}{6}, \qquad \frac{4}{6} + \frac{5}{6} = \frac{9}{6} = \frac{3}{2} = 1\frac{1}{2}$$

Answer: $\frac{3}{2}$ or $1\frac{1}{2}$.

Worked Example: Dividing Fractions

Problem: Evaluate $\frac{3}{4} \div \frac{9}{10}$.
Solution: Multiply by the reciprocal:

$$\frac{3}{4} \times \frac{10}{9} = \frac{30}{36} = \frac{5}{6}$$

Answer: $\frac{5}{6}$.

Worked Example: Mixed Number Multiplication

Problem: Evaluate $2\frac{1}{3} \times 1\frac{2}{5}$.
Solution: Convert to improper fractions first: $2\frac{1}{3} = \frac{7}{3}$, $1\frac{2}{5} = \frac{7}{5}$.

$$\frac{7}{3} \times \frac{7}{5} = \frac{49}{15} = 3\frac{4}{15}$$

Answer: $3\frac{4}{15}$.

3.2.7 Practice Questions

Q1. Evaluate $\frac{5}{8} - \frac{1}{6}$.

Q2. Evaluate $\frac{4}{9} \times \frac{3}{8}$.

Q3. Evaluate $\frac{7}{12} \div \frac{14}{15}$.

Q4. Evaluate $1\frac{3}{4} + 2\frac{1}{6}$.

Q5. A recipe needs $\frac{2}{3}$ cup of sugar per batch. How much sugar for $3\frac{1}{2}$ batches?

Detailed Solutions

A1. LCD of $8, 6$ is 24: $\frac{5}{8} = \frac{15}{24}$, $\frac{1}{6} = \frac{4}{24}$. $\frac{15-4}{24} = \frac{11}{24}$.

A2. $\frac{4 \times 3}{9 \times 8} = \frac{12}{72} = \frac{1}{6}$.

A3. $\frac{7}{12} \times \frac{15}{14} = \frac{105}{168}$. $\mathrm{GCD}(105, 168) = 21$: $\frac{105 \div 21}{168 \div 21} = \frac{5}{8}$.

A4. Convert: $1\frac{3}{4} = \frac{7}{4}$, $2\frac{1}{6} = \frac{13}{6}$. LCD of $4, 6$ is 12: $\frac{7}{4} = \frac{21}{12}$, $\frac{13}{6} = \frac{26}{12}$. Sum: $\frac{47}{12} = 3\frac{11}{12}$.

A5. $\frac{2}{3} \times 3\frac{1}{2} = \frac{2}{3} \times \frac{7}{2} = \frac{14}{6} = \frac{7}{3} = 2\frac{1}{3}$ cups.

3.2.8 Multiple-Choice Questions

MCQ1. $\frac{2}{5} \div \frac{4}{15}$ equals:

(a) $\frac{8}{75}$
(b) $\frac{3}{2}$
(c) $\frac{2}{3}$
(d) $\frac{6}{5}$

Answer: (b) — $\frac{2}{5} \times \frac{15}{4} = \frac{30}{20} = \frac{3}{2}$.

3.2.9 Real-World Applications

Real-World Application: Cooking and Recipe Scaling

Adjusting recipes for different serving sizes requires multiplying every ingredient's fraction by a scaling factor, and adding fractional measurements together when combining ingredients.

Real-World Application: Construction and Carpentry

Measurements in construction (e.g., $\frac{3}{4}$ inch plywood, $2\frac{1}{2}$ inch screws) require precise fraction arithmetic to ensure parts fit together correctly.

3.2.10 Common Mistakes

Common Mistake: Adding Denominators Directly

A very common beginner error: $\frac{1}{3} + \frac{1}{3} = \frac{2}{6}$. This is wrong — denominators are *not* added. The correct approach keeps the common denominator and adds only numerators: $\frac{1}{3} + \frac{1}{3} = \frac{2}{3}$.

Common Mistake: Forgetting to Flip the Second Fraction in Division

Dividing fractions requires multiplying by the reciprocal of the *second* fraction only, not both. $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$, not $\frac{b}{a} \times \frac{d}{c}$.

3.2.11 Interview-Style Conceptual Questions

Conceptual / Interview Question: Why do we need a common denominator to add fractions, but not to multiply them?

Answer: Addition combines like quantities — you can only add "quarters to quarters" or "sixths to sixths," so the parts must be the same size (same denominator) before combining counts. Multiplication, by contrast, represents "a fraction of a fraction" — a scaling operation — so it directly combines numerators and denominators without needing matching part sizes first.

3.2.12 FAQs

FAQ: Do I always need the LCD, or can I use any common denominator?

Any common denominator works and gives a correct (though possibly unsimplified) answer. The LCD is preferred because it keeps the numbers smaller and often avoids an extra simplification step at the end.

Chapter 4

Decimals

4.1 Understanding Decimal Numbers

4.1.1 Beginner Explanation

A **decimal number** uses a decimal point to represent values that are not whole numbers, extending the place-value system to fractional parts. For example, 3.75 means "3 whole units plus 75 hundredths."

Digits to the right of the decimal point represent fractions of powers of ten:

Position	Value
Tenths	$\frac{1}{10}$
Hundredths	$\frac{1}{100}$
Thousandths	$\frac{1}{1000}$

4.1.2 Intermediate Explanation

Every decimal number corresponds to a fraction. **Terminating decimals** (like 0.5 or 0.125) can be written as fractions with a denominator that is a power of 10. **Repeating decimals** (like $0.\overline{3}$) are also rational, and can be converted to fractions using algebraic manipulation. Decimals that neither terminate nor repeat (like $\pi = 3.14159\ldots$) represent irrational numbers.

4.1.3 Advanced Explanation

A fraction $\frac{p}{q}$ in lowest terms has a **terminating** decimal expansion if and only if the prime factorization of q contains only 2s and 5s (since $10 = 2 \times 5$). If q has any other prime factor, the decimal expansion is eventually periodic (repeating) — a consequence of the pigeonhole principle applied to remainders in long division (there are only finitely many possible remainders, so one must eventually repeat).

4.1.4 Key Definitions

Definition: Terminating Decimal

A decimal number with a finite number of digits after the decimal point.

Definition: Repeating (Recurring) Decimal

A decimal in which a digit or block of digits repeats infinitely, denoted with a bar, e.g., $0.\overline{3} = 0.333\ldots$

4.1.5 Formula Sheet

Formula: Decimal–Fraction Conversion

- Terminating decimal to fraction: place digits after the decimal point over the matching power of 10, then simplify. E.g., $0.375 = \frac{375}{1000} = \frac{3}{8}$.
- Repeating decimal $0.\overline{a_1 a_2 \ldots a_n}$ (purely repeating, n digits) $= \dfrac{a_1 a_2 \ldots a_n}{\underbrace{99\ldots9}_{n \text{ nines}}}$

4.1.6 Worked Examples

Worked Example: Converting a Terminating Decimal to a Fraction

Problem: Convert 0.625 to a fraction in lowest terms.
Solution:

$$0.625 = \frac{625}{1000}$$

$\text{GCD}(625, 1000) = 125$:

$$\frac{625 \div 125}{1000 \div 125} = \frac{5}{8}$$

Answer: $\frac{5}{8}$.

Worked Example: Converting a Repeating Decimal to a Fraction

Problem: Convert $0.\overline{45}$ to a fraction.
Solution: Let $x = 0.454545\ldots$. Since two digits repeat, multiply by 100:

$$100x = 45.454545\ldots$$

Subtract the original equation:

$$100x - x = 45.4545\ldots - 0.4545\ldots$$

$$99x = 45 \implies x = \frac{45}{99} = \frac{5}{11}$$

Answer: $\frac{5}{11}$.

Worked Example: Adding Decimals

Problem: Evaluate $12.75 + 3.4 + 0.089$.
Solution: Align decimal points:

$$\begin{array}{r} 12.750 \\ 3.400 \\ +\ 0.089 \\ \hline 16.239 \end{array}$$

Answer: 16.239.

Worked Example: Multiplying Decimals

Problem: Evaluate 2.3×1.4.

Solution: Ignore decimal points and multiply as whole numbers: $23 \times 14 = 322$. Count total decimal places in the factors: $1 + 1 = 2$. Place the decimal point 2 places from the right:

$$2.3 \times 1.4 = 3.22$$

Answer: 3.22.

4.1.7 Practice Questions

Q1. Convert 0.24 to a fraction in lowest terms.

Q2. Convert $0.\overline{7}$ to a fraction.

Q3. Evaluate $45.6 - 12.75$.

Q4. Evaluate 3.5×0.2.

Q5. Evaluate $7.2 \div 0.4$.

Q6. Is $\frac{7}{16}$'s decimal expansion terminating or repeating? Justify using the prime factorization rule.

Detailed Solutions

A1. $0.24 = \frac{24}{100}$; $\text{GCD}(24, 100) = 4$: $\frac{6}{25}$.

A2. Let $x = 0.777\ldots$; $10x = 7.777\ldots$; $10x - x = 7 \Rightarrow 9x = 7 \Rightarrow x = \frac{7}{9}$.

A3. $45.60 - 12.75 = 32.85$.

A4. $35 \times 2 = 70$; total decimal places $= 1 + 1 = 2$; answer $= 0.70 = 0.7$.

A5. $7.2 \div 0.4 = 72 \div 4 = 18$ (multiply both by 10 to clear the decimal in the divisor).

A6. $16 = 2^4$, containing only the prime 2. Since 16's only prime factor is 2, the decimal expansion of $\frac{7}{16}$ **terminates**. (Indeed, $\frac{7}{16} = 0.4375$.)

4.1.8 Real-World Applications

Real-World Application: Currency

Money in most currencies is represented in decimal form (dollars and cents), making decimal arithmetic essential for all financial transactions, taxes, and interest calculations.

Real-World Application: Scientific Measurement

Scientific instruments report measurements as decimals (e.g., 9.81 m/s^2 for gravitational acceleration), and precision in decimal arithmetic directly affects the accuracy of scientific conclusions.

4.1.9 Common Mistakes

Common Mistake: Misaligning Decimal Points When Adding/Subtracting

Decimal points must be vertically aligned before adding or subtracting; otherwise place values get mismatched. Padding with trailing zeros (e.g., writing 3.4 as 3.400) helps avoid this error.

Common Mistake: Miscounting Decimal Places in Multiplication

When multiplying decimals, the number of decimal places in the answer equals the *sum* of decimal places in both factors — not the maximum of the two.

4.1.10 Interview-Style Conceptual Questions

Conceptual / Interview Question: Why do some fractions produce repeating decimals while others terminate?

Answer: It depends on the prime factors of the denominator (in lowest terms). Our number system is base 10, and $10 = 2 \times 5$. If the denominator's only prime factors are 2 and/or 5, the fraction can be rewritten with a denominator that's a power of 10, producing a terminating decimal. Any other prime factor (3, 7, 11, ...) can never be cleared this way, so long division falls into a repeating cycle of remainders.

4.1.11 FAQs

FAQ: Is $0.999\ldots$ actually equal to 1?

Yes. This is a proven mathematical fact, not an approximation. Algebraically: let $x = 0.999\ldots$; then $10x = 9.999\ldots$; subtracting gives $9x = 9$, so $x = 1$. The two decimal representations $0.\overline{9}$ and 1 represent the same real number.

Chapter 5

Percentages

5.1 Understanding Percentages

5.1.1 Beginner Explanation

Percent means "per hundred." A percentage is a way of expressing a number as a fraction of 100. For example, 45% means $\frac{45}{100}$, or 0.45 as a decimal.

5.1.2 Intermediate Explanation

Percentages, fractions, and decimals are three interchangeable representations of the same underlying value:

$$\text{Percent} \longleftrightarrow \text{Fraction} \longleftrightarrow \text{Decimal}$$

$$45\% = \frac{45}{100} = \frac{9}{20} = 0.45$$

To convert a decimal to a percentage, multiply by 100 and add the % sign. To convert a percentage to a decimal, divide by 100 (remove the % sign).

5.1.3 Advanced Explanation

Percentage change calculations combine several core arithmetic ideas (subtraction, division, ratios) and are foundational to compound growth models, which appear throughout finance (compound interest), biology (population growth), and computer science (algorithmic complexity growth rates). Repeated percentage change compounds *multiplicatively*, not additively — a critical distinction covered in §5.1.9.

5.1.4 Key Definitions

Definition: Percent

A ratio expressed as a fraction of 100, denoted with the symbol %.

Definition: Percentage Change

The relative change between an original value and a new value, expressed as a percentage of the original value.

Definition: Percentage Point

A unit representing an absolute difference between two percentages (as opposed to a relative/percentage change). E.g., going from 40% to 50% is a 10 percentage point increase,

but a 25% relative increase.

5.1.5 Formula Sheet

Formula: Percentages

- Percent to decimal: divide by 100
- Decimal to percent: multiply by 100
- Finding a percentage of a value: $\text{Part} = \frac{\text{Percent}}{100} \times \text{Whole}$
- Finding what percent one value is of another: $\text{Percent} = \frac{\text{Part}}{\text{Whole}} \times 100\%$
- Percentage change: $\frac{\text{New} - \text{Original}}{\text{Original}} \times 100\%$
- Finding the whole given a part and percent: $\text{Whole} = \frac{\text{Part}}{\text{Percent}/100}$

5.1.6 Worked Examples

Worked Example: Finding a Percentage of a Number

Problem: What is 35% of 260?
Solution:

$$\frac{35}{100} \times 260 = 0.35 \times 260 = 91$$

Answer: 91.

Worked Example: Finding What Percent One Number Is of Another

Problem: 72 is what percent of 480?
Solution:

$$\frac{72}{480} \times 100\% = 0.15 \times 100\% = 15\%$$

Answer: 15%.

Worked Example: Percentage Increase

Problem: A shirt's price rises from \$40 to \$52. Find the percentage increase.
Solution:

$$\frac{52 - 40}{40} \times 100\% = \frac{12}{40} \times 100\% = 0.3 \times 100\% = 30\%$$

Answer: 30% increase.

Worked Example: Finding the Whole Given a Percentage

Problem: 18 is 24% of what number?
Solution:

$$\text{Whole} = \frac{18}{0.24} = 75$$

Answer: 75.

5.1.7 Practice Questions

Q1. What is 18% of 150?

Q2. 63 is what percent of 180?

Q3. A laptop's price drops from \$900 to \$720. Find the percentage decrease.

Q4. 45 is 30% of what number?

Q5. Convert $\frac{7}{8}$ to a percentage.

Q6. If a population grows from 2000 to 2500, then decreases by 10%, what is the final population?

Detailed Solutions

A1. $0.18 \times 150 = 27$.

A2. $\frac{63}{180} \times 100\% = 0.35 \times 100\% = 35\%$.

A3. $\frac{900-720}{900} \times 100\% = \frac{180}{900} \times 100\% = 20\%$ decrease.

A4. $\frac{45}{0.30} = 150$.

A5. $\frac{7}{8} = 0.875 = 87.5\%$.

A6. First growth: $2000 \to 2500$. Then 10% decrease of 2500: $2500 \times 0.10 = 250$; $2500 - 250 = 2250$. **Final population: 2250.**

5.1.8 Multiple-Choice Questions

MCQ1. If a price increases by 20% and then decreases by 20%, the final price compared to the original is:

(a) Same as original
(b) Higher than original
(c) Lower than original
(d) Cannot be determined

Answer: (c) — increasing by 20% then decreasing by 20% of the *new* (larger) value results in a net decrease. E.g., $100 \to 120 \to 96$.

5.1.9 Real-World Applications

Real-World Application: Discounts and Sales Tax

Retail pricing relies entirely on percentage arithmetic: discounts reduce a price by a percentage, and sales tax adds a percentage. Both require correctly identifying the "whole" (original price) before applying the percentage.

Real-World Application: Compound Interest (Preview)

Percentage change applied repeatedly, as in compound interest, does not add linearly — it compounds multiplicatively. A quantity growing by r% each period for n periods becomes $P(1+\frac{r}{100})^n$, a formula developed fully in the Sequences and Series and Financial Mathematics volumes.

5.1.10 Common Mistakes

Common Mistake: Assuming Percentage Increases and Decreases Cancel Out

As shown in the MCQ above, a 20% increase followed by a 20% decrease does *not* return to the original value, because each percentage is calculated relative to a different base value.

Common Mistake: Confusing Percentage Points with Percent Change

Going from a 10% interest rate to a 12% interest rate is a 2 percentage point increase, but a 20% relative increase ($\frac{2}{10} \times 100\%$). These are frequently confused in news reporting and finance.

5.1.11 Interview-Style Conceptual Questions

Conceptual / Interview Question: Why doesn't a 50% decrease followed by a 50% increase return you to the original value?

Answer: Because each percentage change is computed relative to a different base. A 50% decrease from 100 gives 50. A 50% increase is then computed on the new base of 50, giving $50 + 25 = 75$ — not back to 100. Percentage changes are multiplicative operations on a moving base, not simple additive offsets, which is why sequential percentage changes are asymmetric.

5.1.12 FAQs

FAQ: Can a percentage be greater than 100%?

Yes. Percentages greater than 100% represent values greater than the whole they're compared to, e.g., "sales grew by 150%" means sales more than doubled.

FAQ: Can a percentage be negative?

Yes, a negative percentage typically denotes a decrease, e.g., "-5% growth" means a 5% decline.

Chapter 6

Ratios and Proportions

6.1 Ratios

6.1.1 Beginner Explanation

A **ratio** compares two quantities, showing how much of one thing there is relative to another. It can be written as $a : b$, as a fraction $\frac{a}{b}$, or in words ("a to b"). For example, if a fruit bowl has 3 apples and 2 oranges, the ratio of apples to oranges is $3 : 2$.

6.1.2 Intermediate Explanation

Ratios can be simplified the same way fractions are — by dividing both terms by their GCD. Ratios can also be scaled up or down while preserving the same relationship, and can involve more than two quantities (e.g., $2 : 3 : 5$).

A **proportion** is an equation stating that two ratios are equal: $\frac{a}{b} = \frac{c}{d}$. This is solved using **cross-multiplication**: $ad = bc$.

6.1.3 Advanced Explanation

Ratios and proportions are the foundation of **similarity** in geometry, **scaling** in physics and engineering, and **rates** in calculus (a rate is a ratio of a change in one quantity to a change in another, foreshadowing the derivative). The proportional relationship $y = kx$ (direct proportion, constant ratio k) and $y = \frac{k}{x}$ (inverse proportion, constant product k) are the simplest building blocks of functional relationships, studied further in the Algebra and Precalculus volumes.

6.1.4 Key Definitions

Definition: Ratio

A comparison of two (or more) quantities of the same kind, expressed as $a : b$ or $\frac{a}{b}$.

Definition: Proportion

An equation stating that two ratios are equal: $\frac{a}{b} = \frac{c}{d}$.

Definition: Direct Proportion

A relationship where $y = kx$ for a constant k; as x increases, y increases at a constant rate.

Definition: Inverse Proportion

A relationship where $xy = k$ for a constant k; as x increases, y decreases so that their product stays constant.

6.1.5 Formula Sheet

Formula: Ratios and Proportions

- Simplifying a ratio: divide all terms by their GCD.
- Cross-multiplication: $\dfrac{a}{b} = \dfrac{c}{d} \iff ad = bc$
- Dividing a quantity Q in ratio $a : b$: parts are $\dfrac{a}{a+b}Q$ and $\dfrac{b}{a+b}Q$
- Direct proportion: $y = kx$
- Inverse proportion: $xy = k$, i.e., $y = \dfrac{k}{x}$

6.1.6 Worked Examples

Worked Example: Simplifying a Ratio

Problem: Simplify the ratio $36 : 48$.
Solution: $\mathrm{GCD}(36, 48) = 12$.

$$36 : 48 = 3 : 4$$

Answer: $3 : 4$.

Worked Example: Solving a Proportion

Problem: Solve for x: $\dfrac{x}{15} = \dfrac{4}{5}$.
Solution: Cross-multiply:

$$5x = 15 \times 4 = 60 \implies x = 12$$

Answer: $x = 12$.

Worked Example: Dividing a Quantity in a Given Ratio

Problem: Divide \$720 between two people in the ratio $5 : 7$.
Solution: Total parts $= 5 + 7 = 12$.

$$\text{Person A} = \frac{5}{12} \times 720 = 300, \qquad \text{Person B} = \frac{7}{12} \times 720 = 420$$

Answer: \$300 and \$420. (Check: $300 + 420 = 720$. ✓)

Worked Example: Direct vs. Inverse Proportion

Problem: If 8 workers can build a wall in 15 days, how many days will 12 workers take (assuming constant work rate per worker)?

Solution: More workers means fewer days — this is an **inverse** proportion. Workers $\times$ Days = constant:

$$8 \times 15 = 12 \times d \implies 120 = 12d \implies d = 10$$

Answer: 10 days.

6.1.7 Practice Questions

Q1. Simplify the ratio $54 : 81$.

Q2. Solve for x: $\dfrac{9}{x} = \dfrac{3}{7}$.

Q3. Divide 150 in the ratio $2 : 3 : 5$.

Q4. A map has a scale of $1 : 50000$. If two towns are 3.4 cm apart on the map, what is the actual distance in kilometers?

Q5. If y is directly proportional to x, and $y = 20$ when $x = 4$, find y when $x = 9$.

Q6. If 5 machines take 12 hours to complete a job, how long would 3 machines take (inverse proportion)?

Detailed Solutions

A1. GCD$(54, 81) = 27$: $54 : 81 = 2 : 3$.

A2. Cross-multiply: $9 \times 7 = 3x \implies 63 = 3x \implies x = 21$.

A3. Total parts $= 2 + 3 + 5 = 10$. Shares: $\frac{2}{10} \times 150 = 30$; $\frac{3}{10} \times 150 = 45$; $\frac{5}{10} \times 150 = 75$. Answer: $30, 45, 75$.

A4. Actual distance $= 3.4 \times 50000$ cm $= 170000$ cm $= 1700$ m $= 1.7$ km.

A5. $y = kx \implies 20 = k \times 4 \implies k = 5$. When $x = 9$: $y = 5 \times 9 = 45$.

A6. Constant product: $5 \times 12 = 60$. With 3 machines: $3d = 60 \implies d = 20$ hours.

6.1.8 Real-World Applications

Real-World Application: Maps and Scale Models

Map scales (e.g., $1 : 100000$) and architectural blueprints use ratios to represent real-world distances at a manageable size, and proportions are used to convert between map distance and actual distance.

Real-World Application: Recipe and Mixture Problems

Ratios govern ingredient proportions in cooking, concrete mixing (e.g., cement:sand:gravel ratios), and chemical solutions (e.g., concentration ratios), where getting the ratio wrong can compromise the result.

Real-World Application: Speed, Distance, and Time

The relationship Speed $= \dfrac{\text{Distance}}{\text{Time}}$ is a ratio; at constant speed, distance is directly proportional to time, while at fixed distance, time is inversely proportional to speed.

6.1.9 Common Mistakes

Common Mistake: Confusing Direct and Inverse Proportion

A common error is treating a "more input, less output" situation (like workers and days) as direct proportion. Always ask: as one quantity increases, does the other increase (direct) or decrease (inverse)?

Common Mistake: Not Converting Units Before Comparing Ratios

When forming a ratio between two quantities, both must be in the same units first (e.g., convert cm to m) — otherwise the ratio is meaningless.

6.1.10 Interview-Style Conceptual Questions

Conceptual / Interview Question: How can you tell whether a real-world relationship is directly or inversely proportional?

Answer: Check what happens to one quantity as the other increases while everything else is held constant. If both increase together at a constant rate (their ratio $\frac{y}{x}$ stays constant), it's direct proportion. If one increases while the other decreases such that their product stays constant, it's inverse proportion. A quick sanity check: in direct proportion, doubling x doubles y; in inverse proportion, doubling x halves y.

6.1.11 FAQs

FAQ: Is a ratio the same thing as a fraction?

Closely related but not identical in meaning. A fraction like $\frac{3}{4}$ typically represents "3 parts out of 4 total," while a ratio $3:4$ compares two separate quantities (which together might total 7, not 4). Numerically, however, ratios can be manipulated using fraction arithmetic.

Chapter 7

Simplification of Expressions

7.1 Simplifying Numerical Expressions

7.1.1 Beginner Explanation

Simplification means reducing an expression to its most compact, easy-to-understand form, without changing its value. This chapter brings together everything from Chapters 1–6 — order of operations, fractions, decimals, and ratios — into combined multi-step problems, which is exactly the kind of problem seen on most basic-mathematics exams.

7.1.2 Intermediate Explanation

A reliable strategy for simplifying complex expressions:

1. Simplify inside grouping symbols first (parentheses, brackets, braces), working from the innermost outward.
2. Evaluate exponents and roots.
3. Perform multiplication and division, left to right.
4. Perform addition and subtraction, left to right.
5. Convert mixed numbers to improper fractions if fractions are involved; find common denominators where needed.
6. Simplify the final fraction (if any) to lowest terms.

7.1.3 Advanced Explanation

The BODMAS/PEMDAS convention is really a statement about **operator precedence** — the same underlying concept used in designing programming language parsers and calculators. Expressions with nested grouping symbols are naturally represented as trees, where innermost parentheses correspond to leaves evaluated first — a structure directly connected to how algebraic expressions are later manipulated symbolically in Algebra (Volume 2).

7.1.4 Key Definitions

Definition: Simplification

The process of rewriting a mathematical expression in an equivalent but simpler or more standard form.

Definition: BODMAS/PEMDAS

A mnemonic for order of operations: Brackets/Parentheses, Orders/Exponents, Division & Multiplication (left to right), Addition & Subtraction (left to right).

7.1.5 Formula Sheet

Formula: Simplification Strategy

1. Innermost grouping symbols first
2. Exponents / roots
3. Multiplication / Division (left to right)
4. Addition / Subtraction (left to right)
5. Reduce fractions to lowest terms

7.1.6 Worked Examples

Worked Example: Combined Fraction and Decimal Simplification

Problem: Simplify $\frac{3}{4} + 0.5 \times \left(\frac{2}{3} - \frac{1}{6}\right)$.
Solution: Step 1 — innermost parentheses:

$$\frac{2}{3} - \frac{1}{6} = \frac{4}{6} - \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$

Step 2 — multiplication:

$$0.5 \times \frac{1}{2} = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Step 3 — addition:

$$\frac{3}{4} + \frac{1}{4} = 1$$

Answer: 1.

Worked Example: Nested Grouping Symbols

Problem: Simplify $2 \times \big[3 + \{4 \times (5 - 2)\} \div 6\big]$.
Solution: Innermost first:

$$5 - 2 = 3 \implies 4 \times 3 = 12 \implies 12 \div 6 = 2$$

Then:

$$3 + 2 = 5 \implies 2 \times 5 = 10$$

Answer: 10.

Worked Example: Ratio and Percentage Combined

Problem: A class has students in the ratio of boys to girls $3 : 5$. If there are 40 students total, what percentage are boys?

Solution: Total parts $= 3 + 5 = 8$.

$$\text{Boys} = \frac{3}{8} \times 40 = 15$$

$$\text{Percentage of boys} = \frac{15}{40} \times 100\% = 37.5\%$$

Answer: 37.5%.

7.1.7 Practice Questions

Q1. Simplify $\dfrac{5}{6} \times \left(\dfrac{3}{4} + \dfrac{1}{4}\right) - \dfrac{1}{3}$.

Q2. Simplify $3 \times \big[2 + \{6 \div (4 - 2)\}\big]$.

Q3. Simplify $0.4 \times \left(\dfrac{5}{2} - 1.5\right) + \dfrac{3}{10}$.

Q4. A shop offers a 20% discount, then an additional 10% off the discounted price for members. What single percentage discount does a member effectively receive off the original price?

Q5. Simplify $\left(\dfrac{2}{3}\right)^2 \div \dfrac{4}{9} + 1$.

Detailed Solutions

A1. $\frac{3}{4} + \frac{1}{4} = 1$; $\frac{5}{6} \times 1 = \frac{5}{6}$; $\frac{5}{6} - \frac{1}{3} = \frac{5}{6} - \frac{2}{6} = \frac{4}{6} = \frac{2}{3}$.

A2. $4 - 2 = 2$; $6 \div 2 = 3$; $2 + 3 = 5$; $3 \times 5 = 15$.

A3. $\frac{5}{2} - 1.5 = 2.5 - 1.5 = 1$; $0.4 \times 1 = 0.4$; $0.4 + 0.3 = 0.7$.

A4. Let original price $= 100$. After 20% off: 80. After additional 10% off 80: $80 - 8 = 72$. Effective discount $= 100 - 72 = 28\%$. **Note: this is not simply** $20\% + 10\% = 30\%$ — see Common Mistakes.

A5. $\left(\frac{2}{3}\right)^2 = \frac{4}{9}$; $\frac{4}{9} \div \frac{4}{9} = 1$; $1 + 1 = 2$.

7.1.8 Multiple-Choice Questions

MCQ1. Simplify $\dfrac{1}{2} + \dfrac{1}{3} \times \dfrac{1}{4}$:

(a) $\frac{1}{4}$

(b) $\frac{7}{12}$

(c) $\frac{5}{24}$

(d) $\frac{1}{2}$

Answer: (b) — multiplication first: $\frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$; then $\frac{1}{2} + \frac{1}{12} = \frac{6}{12} + \frac{1}{12} = \frac{7}{12}$.

7.1.9 Real-World Applications

Real-World Application: Successive Discounts and Taxes in Retail

Applying multiple percentage changes in sequence (successive discounts, then tax) requires careful step-by-step simplification, since — as shown above — successive percentages do not simply add together.

7.1.10 Common Mistakes

Common Mistake: Adding Successive Percentage Discounts Directly

A 20% discount followed by a 10% discount is *not* a 30% total discount. Each percentage applies to a different (shrinking) base value. The correct combined effect must be computed step by step, as shown in Practice Question 4 (effective discount here is 28%, not 30%).

Common Mistake: Skipping Steps in Nested Expressions

Attempting to simplify multiple grouping levels at once often causes errors. Always fully resolve the innermost grouping symbol before moving outward.

7.1.11 Interview-Style Conceptual Questions

Conceptual / Interview Question: Why is order of operations described as a "convention" rather than a mathematical law?

Answer: Because the underlying mathematics (the operations themselves) doesn't inherently dictate an order — it's a notational agreement adopted so that any person or system reading the same expression arrives at the same, unambiguous value. Different notations (like Reverse Polish Notation, used in some calculators and programming, or fully parenthesized Lisp-style expressions) avoid the issue entirely by removing ambiguity through structure rather than a precedence convention.

7.1.12 FAQs

FAQ: Should I simplify fractions at every step, or only at the end?

Either approach gives the correct final answer, but simplifying as you go (e.g., reducing $\frac{4}{8}$ to $\frac{1}{2}$ immediately) usually keeps numbers smaller and reduces arithmetic errors in multi-step problems.

Glossary of Mathematical Terms

Absolute Value
: The non-negative distance of a number from zero on the number line, denoted $|x|$.

Composite Number
: A natural number greater than 1 that has a divisor other than 1 and itself.

Decimal
: A number written using place value with a decimal point separating whole and fractional parts.

Denominator
: The bottom number of a fraction, indicating the total number of equal parts.

Direct Proportion
: A relationship $y = kx$ where two quantities increase or decrease together at a constant ratio.

Distributive Property
: The rule $a(b + c) = ab + ac$, allowing multiplication to be distributed over addition.

Divisibility
: The property of one integer being evenly divided by another with no remainder.

Equivalent Fractions
: Different fractions that represent the same numerical value.

Expanded Form
: Writing a number as the sum of the place values of its digits.

Factor (Divisor)
: An integer that divides another integer with no remainder.

Greatest Common Divisor (GCD)
: The largest positive integer that divides two or more numbers without remainder.

Improper Fraction
: A fraction where the numerator is greater than or equal to the denominator.

Integer
: A whole number, its negative, or zero: $\ldots, -2, -1, 0, 1, 2, \ldots$

Inverse Proportion
: A relationship $xy = k$ where one quantity increases as the other decreases, keeping their product constant.

Irrational Number
: A real number that cannot be expressed as a ratio of two integers.

Least Common Multiple (LCM)
The smallest positive integer that is a multiple of two or more given numbers.

Lowest Terms A fraction simplified so that the GCD of its numerator and denominator is 1.

Mixed Number
A number expressed as a whole number combined with a proper fraction.

Multiple A number obtained by multiplying a given number by an integer.

Natural Number
A positive counting number: $1, 2, 3, \ldots$

Numerator The top number of a fraction, indicating how many parts are being considered.

Order of Operations
The agreed sequence for evaluating mathematical expressions (PEMDAS/-BODMAS).

Percent A ratio expressed as a fraction of 100, denoted with the symbol %.

Percentage Change
The relative change between an original and new value, expressed as a percentage of the original.

Place Value The value of a digit based on its position within a number.

Prime Number
A natural number greater than 1 whose only positive divisors are 1 and itself.

Proper Fraction
A fraction where the numerator is less than the denominator.

Proportion An equation stating that two ratios are equal.

Rational Number
A number that can be written as a fraction of two integers with a nonzero denominator.

Ratio A comparison of two or more quantities, written as $a : b$.

Real Number Any number on the number line, encompassing all rational and irrational numbers.

Reciprocal The multiplicative inverse of a number; for $\frac{a}{b}$, the reciprocal is $\frac{b}{a}$.

Repeating Decimal
A decimal number in which one or more digits repeat infinitely.

Simplification The process of reducing an expression to an equivalent but more compact form.

Terminating Decimal
A decimal number with a finite number of digits.

Whole Number
A non-negative integer: $0, 1, 2, 3, \ldots$

Appendices

Formula Appendix

Numbers

- $|x| = x$ if $x \geq 0$; $|x| = -x$ if $x < 0$
- Sum of first n natural numbers: $\dfrac{n(n+1)}{2}$

Divisibility

- $\mathrm{GCD}(a,b) \times \mathrm{LCM}(a,b) = a \times b$
- Euclidean Algorithm: $\mathrm{GCD}(a,b) = \mathrm{GCD}(b, a \bmod b)$

Arithmetic

- Commutative: $a + b = b + a; \quad ab = ba$
- Associative: $(a+b)+c = a+(b+c); \quad (ab)c = a(bc)$
- Distributive: $a(b+c) = ab + ac$
- Order of Operations: Parentheses $\rightarrow$ Exponents $\rightarrow$ Multiplication/Division (L–R) $\rightarrow$ Addition/Subtraction (L–R)

Fractions

- $n\frac{p}{q} = \dfrac{nq+p}{q}$
- $\dfrac{a}{b} + \dfrac{c}{d} = \dfrac{ad+bc}{bd}$
- $\dfrac{a}{b} - \dfrac{c}{d} = \dfrac{ad-bc}{bd}$
- $\dfrac{a}{b} \times \dfrac{c}{d} = \dfrac{ac}{bd}$
- $\dfrac{a}{b} \div \dfrac{c}{d} = \dfrac{ad}{bc}$

Decimals

- Terminating decimal to fraction: digits over matching power of 10, then simplify
- $0.\overline{a_1 \ldots a_n} = \dfrac{a_1 \ldots a_n}{\underbrace{9 \ldots 9}_{n}}$

Percentages

- $\text{Part} = \dfrac{\text{Percent}}{100} \times \text{Whole}$
- $\text{Percent} = \dfrac{\text{Part}}{\text{Whole}} \times 100\%$
- $\text{Percentage change} = \dfrac{\text{New} - \text{Original}}{\text{Original}} \times 100\%$
- $\text{Whole} = \dfrac{\text{Part}}{\text{Percent}/100}$

Ratios and Proportions

- $\dfrac{a}{b} = \dfrac{c}{d} \iff ad = bc$
- Dividing Q in ratio $a : b$: $\dfrac{a}{a+b}Q$ and $\dfrac{b}{a+b}Q$
- Direct proportion: $y = kx$
- Inverse proportion: $xy = k$

Appendices

Symbols Appendix

Symbol	Meaning
$+$	Addition
$-$	Subtraction / negative sign
$\times$ or $\cdot$	Multiplication
$\div$ or $/$	Division
$=$	Equals
$\neq$	Not equal to
$<$	Less than
$>$	Greater than
$\leq$	Less than or equal to
$\geq$	Greater than or equal to
$\|x\|$	Absolute value of x
$\sqrt{x}$	Square root of x
x^n	x raised to the power n
$a : b$	Ratio of a to b
$\frac{a}{b}$	Fraction: a over b
$\%$	Percent
$\mathbb{N}$	Set of natural numbers
$\mathbb{W}$	Set of whole numbers
$\mathbb{Z}$	Set of integers
$\mathbb{Q}$	Set of rational numbers
$\mathbb{R}$	Set of real numbers
$\mathbb{C}$	Set of complex numbers
$\in$	"Is an element of"
$\subset$	"Is a proper subset of"
$\text{GCD}(a, b)$	Greatest common divisor of a and b
$\text{LCM}(a, b)$	Least common multiple of a and b
$0.\overline{n}$	Repeating decimal, digit(s) n repeat infinitely
π	Pi, approximately 3.14159